

Culminating Activity

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Abstract

Provide a concise summary of the study including purpose, methods, results, and conclusions (150–250 words).

Chapter 1: Introduction

Background of the Study

Global sales prediction plays an important role in helping businesses make informed decisions related to marketing strategies, inventory management, budgeting, and future business planning. Many organizations rely on predictive models such as Multiple Linear Regression to analyze trends and forecast sales performance using historical and market-related data. However, one of the common challenges in predictive analytics is the presence of missing data within datasets. Missing values may occur due to incomplete records, human error, system failures, or inconsistent data collection processes. When these missing values are not properly handled, they can negatively affect the accuracy, consistency, and reliability of sales prediction models, leading to poor decision-making and inaccurate forecasts.

This research focuses on examining the effect of missing value handling on improving global sales prediction using Multiple Linear Regression. Specifically, it aims to understand why missing data reduces model performance and how different data cleaning techniques can enhance predictive accuracy. It also highlights the significance of proper data preparation in machine learning and statistical analysis. By identifying effective preprocessing methods, the research can help businesses, researchers, and data analysts develop more reliable prediction models and improve the quality of data-driven decisions in global sales forecasting.

Statement of the Problem

1. What is the effect of missing value handling on improving global sales prediction using Multiple Linear Regression?
2. Why does missing data reduce the accuracy and reliability of sales prediction models?
3. How does applying data cleaning techniques improve the performance of the Multiple Linear Regression model?

Objectives of the Study

General Objective

- To investigate the effect of missing value handling on improving global sales prediction using Multiple Linear Regression.

Specific Objectives

- To determine how missing data affects the accuracy and reliability of sales prediction models.
- To implement and assess data imputation strategies to prevent the loss of valuable data caused by deleting incomplete rows.
- To compare the performance metrics between models using data with deleted missing rows and models using imputed data.

Significance of the Study

Explain who benefits and how.

Scope and Limitations

This research focuses on predicting global video game sales by comparing two data-handling methods which are deleting missing rows and using the Bayesian Linear Regression imputation. The analysis is specifically performed using Multiple Linear Regression within the R programming environment. The primary goal is to determine which data-handling technique leads to a more accurate and reliable prediction model.

Chapter 2: Review of Related Literature

Related Studies

In a study focused specifically on video game sales data from Kaggle, Affan et al. (2024) developed a machine learning-based sales prediction model using several regression algorithms. Their model made use of regression algorithms, including Linear Regression, Ridge Regression, and Gradient Boosting Regressor, to forecast worldwide video game sales based on variables such as genre, platform, publishers, critic reviews, and user ratings. Their work demonstrates that the same video game sales dataset used in this study has been applied in prior research, validating its academic relevance.

Separately, a study on predicting video game sales using supervised learning from the same Kaggle dataset highlighted the impact of missing data on model outcomes. After downloading the dataset, the researcher dropped rows with empty values or filled in missing values to standardize the columns before applying regression modeling, noting that the dataset contained missing score fields for some video games, which posed challenges to prediction reliability. This finding directly parallels the core concern of this research, that unhandled missing data undermines prediction accuracy. Du (2025).

Ahmad et al. (2024) conducted a comprehensive review of machine learning applications specifically for missing value imputation. Their analysis sought to enhance researchers' understanding of MVI methods and facilitate the development of robust interventions in data preprocessing for data analytics, elucidating the strengths and limitations of different ML algorithms in imputing missing values across different types of data. This reinforces the idea that the choice of imputation technique has measurable consequences for downstream model performance.

Related Literature

Multiple Linear Regression

Multiple Linear Regression is a fundamental statistical technique used to model the relationship between one dependent variable and two or more independent variables. According to Fiveable (2024), the goal of

MLR is to find the line of best fit that minimizes the sum of squared residuals, where the model takes the form $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$, with each slope coefficient representing the change in the dependent variable for a one-unit change in the corresponding independent variable, holding all others constant.

The reliability of MLR estimates depends on a set of classical assumptions known as the Gauss-Markov conditions. As explained by Jim (n.d.), the Gauss-Markov theorem states that if a linear regression model satisfies the classical assumptions, then Ordinary Least Squares (OLS) regression produces unbiased estimates that have the smallest variance of all possible linear estimators. When these assumptions are violated, for instance, due to missing data introducing bias or inconsistency into the dataset, the estimates produced by the model can no longer be guaranteed to be optimal.

Missing Data Mechanisms

The theoretical foundation for understanding missing data was established by Rubin (1976) and later expanded upon by Little and Rubin. As described by the Statistical Consulting Centre of the University of Melbourne (n.d.), data is considered Missing Completely at Random (MCAR) if the probability of being missing does not depend on the values of any of the variables in the data, Missing at Random (MAR) if the probability of being missing depends only on observed data, and Missing Not at Random (MNAR) if the probability of missingness is related to the values that would have been observed had the data not been missing. Understanding which mechanism governs the missing values in a dataset is critical because it determines which imputation strategies are appropriate and what biases might be introduced if missing data is simply ignored.

Further elaborated by APXML (n.d.), the MCAR assumption is the simplest scenario, where missingness is purely random and does not have a systematic cause related to the data, while MAR is often a reasonable working assumption when MCAR seems too simplistic and there is no strong domain reason to suspect MNAR. According to ScienceDirect (n.d.), the use of inappropriate methods for handling missing data can lead to bias in parameter estimates, bias in standard errors and test statistics, and inefficient use of data, all of which directly degrade the predictive accuracy of a regression model.

Data Preprocessing and Model Performance

Data preprocessing refers to a set of techniques applied to raw data before it is used for analysis or model training. As highlighted by Lee (2025), simple steps like normalization and feature scaling can have dramatic effects on model performance, and only including features that contribute significantly to prediction accuracy is essential to building a reliable regression model. In the context of sales prediction, preprocessing steps such as handling missing values, removing duplicate records, and encoding categorical variables are necessary to ensure that the dataset meets the assumptions required for valid regression analysis.

Supporting this, research cited in the IJEDR (2025) by Mehta and Sharma (2022) demonstrated that removing outliers and cleaning data improved model performance significantly, while Singh and Gupta (2022) compared various regression models and found that proper preprocessing influenced predictive results across model types. These findings collectively support the rationale of this study: that systematic application of data cleaning techniques prior to fitting a Multiple Linear Regression model will yield measurably improved performance in global sales prediction.

Conceptual Framework

(Insert diagram if needed)

Chapter 3: Methodology

Research Design

Describe the design (e.g., descriptive, experimental).

Data Collection

The dataset utilized in this study was collected from Kaggle, <https://www.kaggle.com/code/yonatanrabinovich/video-games-sales-regression-techniques/input>.

```
# Load dataset
data <- read.csv("../Video_Games_Sales_as_at_22_Dec_2016.csv", sep=";", na.strings=c("", " ", "NA", "N/A"))
head(data)
```

```
##           Name Platform Year_of_Release      Genre Publisher
## 1      Wii Sports      Wii          2006      Sports  Nintendo
## 2    Super Mario Bros.    NES          1985 Platform  Nintendo
## 3      Mario Kart Wii      Wii          2008      Racing  Nintendo
## 4    Wii Sports Resort      Wii          2009      Sports  Nintendo
## 5 Pokemon Red/Pokemon Blue  GB          1996 Role-Playing Nintendo
## 6          Tetris        GB          1989      Puzzle  Nintendo
##   NA_Sales EU_Sales JP_Sales Other_Sales Global_Sales Critic_Score Critic_Count
## 1    41.36   28.96    3.77      8.45      82.53          76           51
## 2    29.08    3.58    6.81      0.77      40.24          NA           NA
## 3    15.68   12.76    3.79      3.29      35.52          82           73
## 4    15.61   10.93    3.28      2.95      32.77          80           73
## 5     11.27    8.89   10.22      1.00      31.37          NA           NA
## 6     23.20    2.26    4.22      0.58      30.26          NA           NA
##   User_Score User_Count Developer Rating
## 1          8        322   Nintendo      E
## 2         <NA>         NA      <NA>    <NA>
## 3          8.3       709   Nintendo      E
## 4          8        192   Nintendo      E
## 5         <NA>         NA      <NA>    <NA>
## 6         <NA>         NA      <NA>    <NA>
```

Variables Description

- Global_Sales (Dependent Variable)
- Platform (Independent Variable)
- Game_Age (Independent Variable)
- Genre (Independent Variable)
- Critic_Score (Independent Variable)
- Critic_Count (Independent Variable)
- User_Count (Independent Variable)
- Rating (Independent Variable)

Explain dependent and independent variables.

Data Analysis Techniques

- Descriptive Statistics
- Inferential Statistics
- Machine Learning Models (if applicable)

Chapter 4: Results and Discussion

Descriptive Statistics

```
##      Name           Platform      Year_of_Release      Genre
## Length:16719      Length:16719      Min.   :1980      Length:16719
## Class :character  Class :character  1st Qu.:2003      Class :character
## Mode  :character  Mode  :character  Median :2007      Mode  :character
##                                     Mean  :2006
##                                     3rd Qu.:2010
##                                     Max.   :2020
##                                     NA's   :269
## Publisher          NA_Sales          EU_Sales          JP_Sales
## Length:16719      Min.   : 0.0000      Min.   : 0.000      Min.   : 0.0000
## Class :character  1st Qu.: 0.0000      1st Qu.: 0.000      1st Qu.: 0.0000
## Mode  :character  Median : 0.0800      Median : 0.020      Median : 0.0000
##                                     Mean  : 0.2633      Mean  : 0.145      Mean  : 0.0776
##                                     3rd Qu.: 0.2400      3rd Qu.: 0.110      3rd Qu.: 0.0400
##                                     Max.   :41.3600      Max.   :28.960      Max.   :10.2200
##
## Other_Sales        Global_Sales        Critic_Score        Critic_Count
## Min.   : 0.00000      Min.   : 0.0100      Min.   :13.00      Min.   : 3.00
## 1st Qu.: 0.00000      1st Qu.: 0.0600      1st Qu.:60.00      1st Qu.: 12.00
## Median : 0.01000      Median : 0.1700      Median :71.00      Median : 21.00
## Mean   : 0.04733      Mean   : 0.5335      Mean   :68.97      Mean   : 26.36
## 3rd Qu.: 0.03000      3rd Qu.: 0.4700      3rd Qu.:79.00      3rd Qu.: 36.00
## Max.   :10.57000      Max.   :82.5300      Max.   :98.00      Max.   :113.00
##                                     NA's   :8582      NA's   :8582
## User_Score          User_Count          Developer          Rating
## Length:16719      Min.   : 4.0      Length:16719      Length:16719
## Class :character  1st Qu.: 10.0      Class :character  Class :character
## Mode  :character  Median : 24.0      Mode  :character  Mode  :character
##                                     Mean   : 162.2
##                                     3rd Qu.: 81.0
##                                     Max.   :10665.0
##                                     NA's   :9129
```

Visualization

```
library(ggplot2)
#ggplot(data, aes(x=Variable1, y=Variable2)) +
#  geom_point() +
#  theme_minimal()
```

Model

```
#model <- lm(total_sales~., numeric_data)
#summary(model)
```

Interpretation

Discuss findings in paragraph form.

Chapter 5: Conclusion and Recommendations

Conclusion

Summarize key findings.

Recommendations

Provide practical suggestions

References

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Appendices

Appendix A: Code